

Random Search for Treasure With Sure Chance of Success

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Background and Motivation

Cops and Robbers

- The original motivation of this project came from cops and robbers.
- The case where the cop does not know where the robber is seemed interesting to me and was something I wanted to explore.

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Drunken cop and tipsy robber

- This is a version of cops and robbers where the cop and robber both have a certain probability of moving randomly.
- I thought this could be improved by making the cop always move randomly.

Random Movement of cop

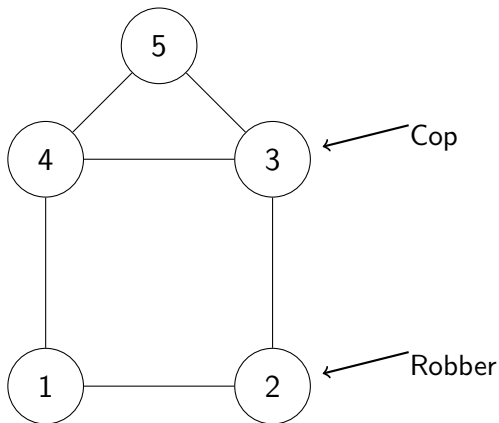
Movement of cop

- The cop has a random probability of moving to any of the vertices in the neighborhood of the current vertex it is on
- After the cop moves from a vertex, the neighborhood of that vertex becomes a complete subgraph

Manipulation of graph

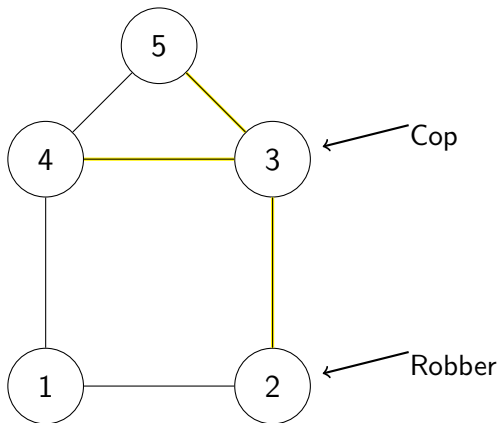
- The graph gets smaller each time the cop moves
- The vertex the cop was previously on gets deleted and a complete subgraph is created from its neighborhood
- This allows the cop to essentially "fast travel" through vertices it has already traveled to

Example: House Graph



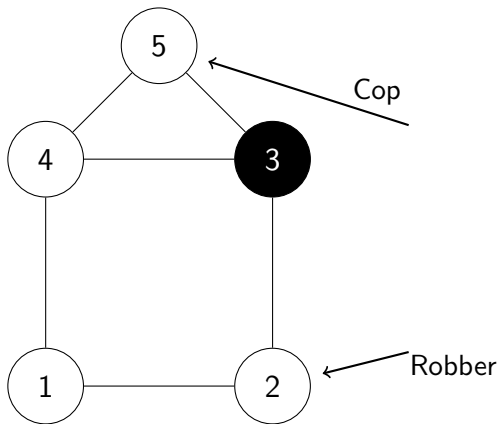
Robber is starting at vertex 2 and cop is starting at vertex 3.

Example: House Graph



Cop can travel to vertex 2, 4, or 5

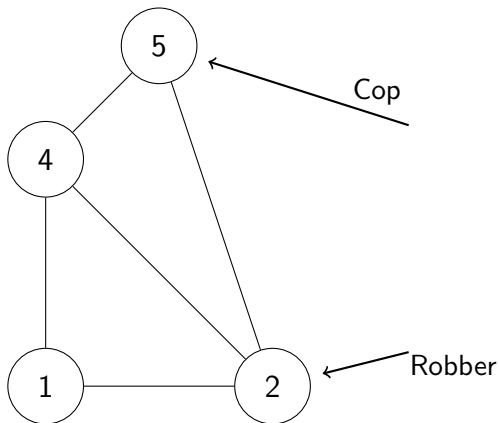
Example: House Graph



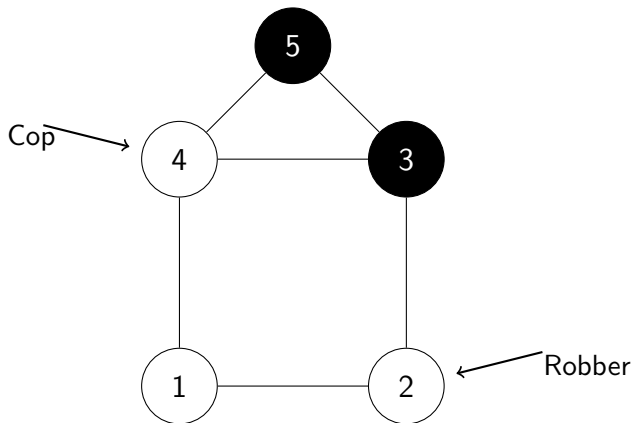
Turns: 1

Probability: $\frac{1}{3}$

Example: House Graph



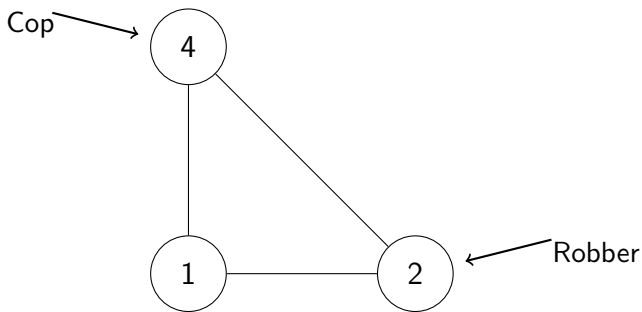
Example: House Graph



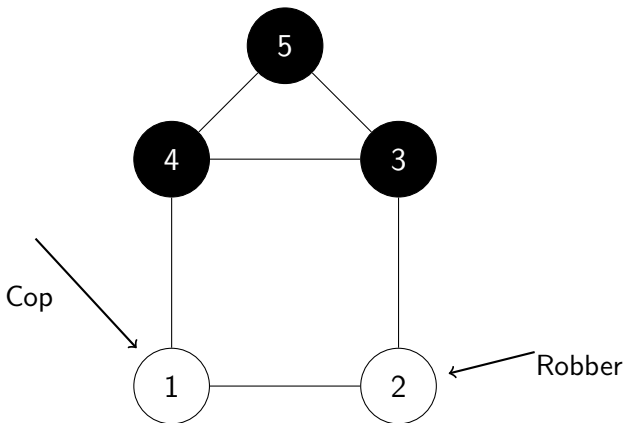
Turns: 2

Probability: $\frac{1}{3} * \frac{1}{2}$

Example: House Graph



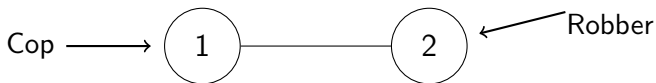
Example: House Graph



Turns: 3

Probability: $\frac{1}{3} * \frac{1}{2} * \frac{1}{2}$

Example: House Graph



Turns: 4

$$\text{Probability: } \frac{1}{3} * \frac{1}{2} * \frac{1}{2} = \frac{1}{12}$$

$$\text{Probability Density: } \frac{1}{3}$$

Robber Starting Position

- The robber wants to avoid capture for as long as possible
- For each cop starting position, The robber start time will be the maximum expected capture time
- There will be $|V|$ of these maximum capture times

Cop Starting Position

- The cop wants to capture the robber as fast as possible
- The cop will start where the minimum capture time of the $|V|$ maximum capture times is.

The Problem

- Finding the cop and robber starting positions by hand is very tedious, especially on big graphs
- This is a perfect case for a recursive algorithm

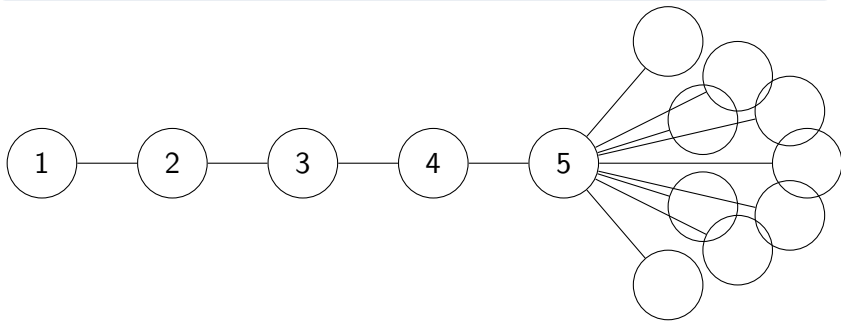
The solution

I created an algorithm that loops through all possible cop and robber starting positions, outputting the average capture time at each one

Goals and Next Steps

Finding Cop and Robber Starting Positions

- I would like to find a formula that finds cop and robber starting positions
- Using the simulation I created we can easily see the starting cop and robber positions or any relatively small graph
- The starting position is not necessarily always the center of the graph



Goals and Next Steps

Questions

- Look at other ways of contracting the graph
- Let the robber move, either randomly or non-randomly
- Add an additional randomly moving cop
- Combine this idea with edge burning robbers